

AURA

Product & Technical Specification

MVP: Musée d'Orsay Edition

Version 1.1 · 2 August 2026 · Codex-ready

POINT. DISCOVER. UNDERSTAND.

Point your camera at an artwork. Discover its story, significance and scale. After the museum, see how many masterpieces — and billions — you experienced.

1. Executive summary

AURA is a game-like AI museum guide. A visitor points the camera at an artwork and, within seconds, receives a compact, understandable card: what the object is, why it matters, what to look at, how rare it is, and an honest indicative market-value range. During the visit, AURA tracks works, artists, time, progress, missions, and total indicative value. At the end, it creates a vertical share card for Instagram Stories, TikTok, WhatsApp, and image export.

AURA is not an encyclopedia, a long audio guide, or an art-history course. The first release is deliberately narrow: one museum, a highly reliable recognition system, one excellent artwork card, and one viral visit recap.

2. Product thesis

Museums usually present art through labels: artist, year, technique and dimensions. Most visitors cannot translate those facts into meaning. AURA turns a label into a story that feels native to the Instagram/TikTok era without reducing art to empty entertainment.

- Hook — scale: “What might this be worth?”
- Utility — meaning: “Why does this matter?”
- Emotion — attention: “Where should I look to feel it myself?”
- Reward — recap: “Today you experienced 37 masterpieces with an indicative value of €3.8B.”

Money is a translator of scale, not the meaning of the artwork. The primary product is understanding.

3. MVP goals and success criteria

1. Visitors use AURA’s camera instead of the default phone camera. Target: more than 60% of a test group perform at least five scans.
2. Visitors read the card and return their attention to the physical artwork. Target: median card engagement above 15 seconds and measurable increase in artwork dwell time.
3. Visitors share the recap. Target: 30% of completed visits trigger a share or image save.

North-star metric: Increase in meaningful attention time in front of the original artwork.

4. MVP scope

4.1 Included

- Musée d’Orsay only.
- Top 100 priority works recognized; Top 20 fully editorialized at launch.
- Retrieval-based AI visual recognition against a controlled reference set.
- Artwork card: identity, indicative value, analogy, significance, “where to look,” rarity and 30–60 second audio.
- Visit session, progress, missions, favorites and recap.
- English, French and Simplified Chinese from day one.
- Normal, Simple and Kids explanation modes.
- Analytics required to validate the three MVP hypotheses.
- Monetization architecture prepared but activated only after validation.

4.2 Explicitly excluded

- Global Cultural Score, rankings or leaderboards.
- Cultural biography and AI personality profile.
- Social graph and public feed.

- AR historical reconstructions.
- B2B dashboards, heatmaps and white label.
- Multiple museums in the launch build.
- Generic “recognize all art in the world” positioning.
- Unverified AI-generated facts or live AI-generated valuations.

5. Target users

Primary: international visitors aged 18–55 who use mobile photography and social media, are curious about art, but do not want a traditional academic guide.

Secondary: parents with children, young adults, visitors interested in collecting and value, and French-, English- and Chinese-speaking tourists in Paris.

6. Core user journey

4. Open AURA.
5. Detect Musée d’Orsay by GPS/geofence or choose it manually.
6. Select EN, FR or 简体中文; suggest the device language.
7. Tap Start Visit and receive three missions.
8. Photograph an artwork.
9. Recognize and return a localized card in under 2.5 seconds for known works.
10. Read, listen, add, favorite or share.
11. Update progress automatically.
12. Complete the visit and generate a 9:16 recap.

7. Screen specification

The MVP contains six user-facing states. They may be implemented as five main screens plus onboarding.

7.1 Onboarding

One-screen explanation, language selector EN / FR / 简体中文, and Continue CTA.

7.2 Museum Home

Museum detection chip, Start/Continue Visit, three rotating missions and a short “How it works” row.

7.3 Camera

Fullscreen camera, 3:4 framing guide, haptic shutter, card skeleton while loading, and a calm retry state.

7.4 Artwork Card

Image, artist/title/year, estimate badge and disclaimer, analogy, Why it matters, Normal/Simple/Kids mode, Where to look, rarity, audio, Add/Favorite/Share.

7.5 Visit Progress

Works, artists, value, duration, route completion, missions, viewed gallery and next action.

7.6 Visit Recap

1080×1920 PNG with museum, metrics, most valuable work, favorite, badge and share sheet.

Artwork card example

Artwork	Claude Monet — Vétheuil, soleil couchant — 1901
---------	---

Estimate	€80–120M · indicative, not a sale or insurance value
Why it matters	Monet stopped describing the city precisely and began showing how light changes it from moment to moment.
Where to look	Step back three paces and watch separate brushstrokes become a luminous reflection.
Rarity	Works of this level almost never reach the open market.
Actions	Add to Visit · Listen · Favorite · Share

8. AI and recognition architecture

8.1 Principle

Recognition is primarily retrieval, not unconstrained generation. Reliability on a controlled museum set is more important than worldwide breadth: 98% accuracy on 100 works is better than 60% on 100,000.

8.2 Pipeline

13. Capture and normalize image on device to 512×512.
14. Send image with museum_id and optional hall_hint.
15. Generate DINOv2 or CLIP ViT-L/14 embedding.
16. Search FAISS or Supabase Vector.
17. Re-rank using museum, hall, aspect ratio, optional wall-label OCR and optional multimodal verification.
18. Return artwork_id, confidence and alternatives.

8.3 Confidence policy

- ≥ 0.92 : auto-confirm.
- $0.82\text{--}0.92$: ask user to confirm top candidates or run verification.
- < 0.82 : request a new photo.

The UI must never confidently present an uncertain match.

8.4 Generative AI boundaries

- May draft and translate editorial copy.
- May adapt verified meaning into Normal, Simple and Kids modes.
- May generate audio from approved scripts.
- Must not invent identity facts, rarity, comparable sales or valuations.
- Valuations are stored and reviewed, never generated live.

9. Data architecture

9.1 Two-layer model

Layer 1 — Factual DB (approximately 80%, imported): artist, titles, year, technique, dimensions, museum, inventory, hall, public-domain image, sources and tags.

Layer 2 — AURA editorial content (approximately 20%, reviewed): estimate, logic, comparable sales, analogy, significance, observation instruction, rarity, audio, missions and editorial confidence.

9.2 Canonical entities

- museums
- artworks

- artwork_localizations
- artwork_estimates
- artwork_embeddings
- visits
- visit_artworks
- missions

The spreadsheet is a CMS import/export surface. Production data must be normalized in the database.

10. Multilingual and localization requirements

Launch locales: English (en), French (fr), Simplified Chinese (zh-Hans). The architecture must allow additional locales without schema changes.

- No user-facing string is hard-coded in views.
- UI uses localization keys; artwork editorial content is stored by locale.
- Preserve original artwork titles and provide localized titles.
- Fallback: requested locale → English → original factual field.
- EUR estimates use locale-aware number formatting.
- Chinese must be naturally localized in Simplified Chinese, not literally machine-translated.
- English is the canonical fallback.

Core localized UI strings

Key	English	French	简体中文
start_visit	Start Visit	Commencer la visite	开始参观
frame_artwork	Frame the entire artwork	Cadrez l'œuvre en entier	请将整件作品置于画面中
add_to_visit	Add to Visit	Ajouter à ma visite	加入本次参观
why_it_matters	Why it matters	Pourquoi cette œuvre compte	为什么它重要
where_to_look	Where to look	Où regarder	看哪里
indicative_estimate	Indicative market estimate	Estimation indicative de marché	市场参考估值
share_visit	Share your visit	Partager votre visite	分享本次参观
try_again	Try again	Réessayer	重试

Audio

- Normal audio in EN, FR and zh-Hans for Top 20.
- Separate reviewed script per locale.
- 30–60 seconds; natural duration, not literal word-for-word parity.
- Cache TTS output; do not synthesize on every play.

11. Content rules

Estimates

Always show a range. Mandatory disclaimer: This museum work is not for sale. The range is an editorial estimate based on comparable public sales, artist, period, subject, size, provenance and museum significance. It is not an appraisal or insurance value.

Analogies

Use a controlled, dated reference table and approximate language such as “Comparable to...”. Avoid false precision.

Why it matters

One sentence, no academic jargon, explaining the change represented by the work.

Where to look

One direct action beginning with a verb and sending the user's eyes back to the physical artwork.

Audio

A calm, knowledgeable friend — not a lecturer. Combine one meaning with one observation action.

12. API contract

POST /v1/recognize

```
{
  "image_base64": "...",
  "museum_id": "orsay",
  "hall_hint": "34",
  "locale": "fr"
}

→ {
  "status": "matched",
  "artwork_id": "orsay_rf_1990",
  "confidence": 0.96,
  "alternatives": []
}
```

- GET /v1/artworks/{id}?locale=en&mode=normal
- POST /v1/visits
- POST /v1/visits/{visit_id}/artworks
- GET /v1/visits/{visit_id}/progress
- POST /v1/visits/{visit_id}/complete

13. Technical stack

Mobile

- iOS-first: SwiftUI.
- Camera: AVFoundation.
- Location: CoreLocation.
- Localization: String Catalog / .xcstrings.
- Anonymous first visit; account not required.

Backend

- Python FastAPI.
- Supabase: Postgres, Storage, optional pgvector.
- FAISS in memory is acceptable for Top 100.
- Background jobs for imports, embeddings, audio and share-card rendering.

Analytics

Required events: onboarding_completed, language_selected, visit_started, scan_attempt, scan_success, scan_failed, candidate_confirmed, artwork_card_opened, artwork_card_read_time, audio_started, audio_completed, artwork_added, artwork_favorited, mission_completed, visit_completed, recap_generated, share_started, share_completed, paywall_viewed, purchase_completed.

14. Monetization — prepared, activated later

Free

Five recognitions, basic identification and short explanation, recap without value total.

Museum Pass — €4.99

Unlimited recognition for 24 hours in Orsay; full cards, values, audio, missions, Kids mode and full recap.

Paris Pass — €12.99

Future Paris museums. Do not sell unavailable content.

Monetization is implemented only after behavior validation.

15. Design direction

- Apple-first, contemporary and calm.
- References: Apple Camera, Shazam, Linear and Airbnb.
- Light theme: #FAFAF9 background, #111111 text, black primary button.
- Approximately 20 px corner radius; SF Pro on iOS, Inter for admin tooling.
- Generous whitespace; no museum gold, parchment or faux-classical clichés.
- Price hooks attention but must not visually dominate meaning and observation.

16. CMS workbook mapping

The two uploaded workbook variants are consolidated into one canonical editorial import model. Localized text is wide in the spreadsheet for editing, then normalized into artwork_localizations on import.

- ID
- Museum ID
- Artist
- Title Original
- Title EN
- Title FR
- Title zh-Hans
- Year
- Inventory Number
- Hall
- Technique
- Dimensions
- Image URL
- Priority
- Estimate Low EUR M
- Estimate High EUR M
- Estimate Logic
- Comparable Sales
- Estimate Confidence
- Analogy Type
- Analogy EN/FR/zh-Hans
- Analogy Kids EN/FR/zh-Hans
- Why Important Normal EN/FR/zh-Hans
- Why Important Simple EN/FR/zh-Hans

- Why Important Kids EN/FR/zh-Hans
- Where to Look EN/FR/zh-Hans
- Rarity Note EN/FR/zh-Hans
- Audio Script EN/FR/zh-Hans
- Tags
- Mission IDs
- Source URLs
- Editorial Status
- Reviewed By
- Updated At

17. Delivery plan

Week 1 — Data and recognition

- Finalize Top 100 factual records.
- Build reference-image set and embeddings.
- Implement /recognize and confidence handling.
- Create DB schema and localization keys.
- Import workbook into staging tables.

Week 2 — iOS product flow

- Onboarding and language selection.
- Museum Home.
- Camera and recognition states.
- Artwork Card.
- Visit Progress.
- Localized UI in EN, FR and zh-Hans.

Week 3 — Content, recap and field test

- Review Top 20 content in three languages.
- Generate and cache audio.
- Implement Recap and PNG export.
- Add analytics.
- TestFlight field test inside Musée d'Orsay.
- Measure accuracy, card reading, dwell time and sharing.

This is an aggressive three-week plan. Scope must remain frozen.

18. Definition of done

- Top 100 known works recognized under normal museum conditions.
- Top 20 have reviewed EN, FR and zh-Hans cards and audio.
- Recognition p50 <2.5s and p95 <5s on 4G.
- Low-confidence matches never produce confident false results.
- Anonymous visit starts, records works, updates missions and completes.
- Recap exports correctly at 1080×1920.
- No raw localization keys appear.
- All estimates include disclaimer.
- Analytics works end to end.
- One real-day Musée d'Orsay pilot completed.

19. Codex implementation order

19. Repository and environment configuration.
20. Database migrations and seed format.
21. Localization framework with EN, FR and zh-Hans from day one.
22. Workbook import script.
23. Recognition service with local test dataset.
24. Typed API contracts and automated tests.
25. iOS onboarding, home and visit state.
26. Camera and recognition flow.
27. Artwork Card and content modes.
28. Visit Progress and missions.
29. Recap rendering and sharing.
30. Analytics.
31. Audio.
32. Monetization only after pilot review.

Engineering rules

- Do not fabricate artwork facts.
- Do not generate valuations at request time.
- Do not hard-code user-facing strings.
- Do not couple UI directly to spreadsheet columns.
- Every endpoint returns typed errors.
- Every recognition result includes confidence.
- Prefer simple, observable systems over premature scale architecture.

20. Positioning

Consumer: Point. Discover. Understand.

French: Cadrez. Découvrez. Comprenez.

Chinese: 对准作品，发现故事，读懂艺术。

Internal/investor: AURA is an AI-powered, game-like museum guide that turns physical artworks into understandable stories, scale and shareable visit achievements.

21. Final product principle

Do not build a universal cultural operating system yet. Build one moment exceptionally well:

A visitor photographs a Monet in Musée d'Orsay, receives a trustworthy and beautiful explanation in their language, looks back at the original with new attention, says "wow," and shares the visit with friends.

If that loop works, the larger AURA can be built on top of it.